HW4

313551170 王韋紀

1. Logistic Regression

$$D = \begin{bmatrix} x_1 & y_1 \\ x_2 & y_2 \\ \vdots & \\ x_{2n} & y_{2n} \end{bmatrix} \quad (D_1 + D_2) \quad , N = 2n$$

$$z = ax + by + c = w_0 + w_1 x + w_2 y$$

$$w = [w_0, w_1, w_2]^T, \quad \Phi = [1, x, y]^T, \quad z = w^T \Phi$$

$$\text{label} = y' = [\overbrace{0, 0, \ldots, 0}^{n}, \overbrace{1, \ldots, 1}^{n}]^T, \quad 0 \text{ for } D_1, \quad 1 \text{ for } D_2$$

$$\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-w^T \Phi}}$$

Loss: (Binary Cross Entropy)

$$J(w) = -\frac{1}{N} \sum_{i=1}^{N} [y_i' \ln \hat{y}_i + (1-y_i') \ln (1-\hat{y}_i)]$$

$$= -\frac{1}{N} \sum [y_i' \ln \sigma(z) + (1-y_i') \ln (1-\sigma(z))]$$

$$\frac{\partial \sigma(z)}{\partial z} = \sigma(z)(1-\sigma(z)) = \hat{y}(1-\hat{y})$$

$$\frac{\partial J(w)}{\partial z} = \frac{\partial J(w)}{\partial \sigma(z)} \cdot \frac{\partial \sigma(z)}{\partial z} = -\frac{1}{N} \sum \left(\frac{y'}{\hat{y}} - \frac{1-y'}{1-\hat{y}} \right) \cdot \hat{y}(1-\hat{y})$$

$$= -\frac{1}{N} \sum [y'(1-\hat{y}) - \hat{y}(1-y')]$$

$$= -\frac{1}{N} \sum (y' - \hat{y})$$

$$= \frac{1}{N} \sum (\hat{y} - y')$$

$$\frac{\partial J(w)}{\partial w_0} = \frac{\partial J(w)}{\partial z} \frac{\partial z}{\partial w_0}$$

$$= \frac{1}{N} \sum (\hat{y}_i - y'_i)$$

$$\frac{\partial J(w)}{\partial w_1} = \frac{1}{N} \sum (\hat{y}_i - y'_i) x_i$$

$$\frac{\partial J(w)}{\partial w_2} = \frac{1}{N} \sum (\hat{y}_i - y'_i) y_i$$

- Gradient descent:

Gradient:

$$w' = \begin{bmatrix} w_0' \\ w_1' \\ w_2' \end{bmatrix} = \frac{1}{N} \begin{bmatrix} 1 & \cdots & 1 \\ x_1 & \cdots & x_N \\ y_1 & \cdots & y_N \end{bmatrix} \left(\begin{bmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_N \end{bmatrix} - \begin{bmatrix} y'_1 \\ \vdots \\ y'_N \end{bmatrix} \right)$$

實際運算:

$$\Phi = \begin{bmatrix} 1 & \cdots & 1 \\ x_1 & \cdots & x_N \\ y_1 & \cdots & y_N \end{bmatrix}, \quad 3 \times N, \quad N = 2n$$

$$y' = \begin{bmatrix} y'_1 \\ \vdots \\ y'_N \end{bmatrix}, \quad w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

$$z = \Phi^T w$$

$$\hat{y} = \sigma(z)$$

$$w' = \frac{1}{N} \Phi (\hat{y} - y')$$

$$w_{new} = w_{old} - \alpha w'$$

- Newton's method

$$W_{new} = W_{old} + \frac{-J'(W_{old})}{H(J(W_{old}))}$$

$$J'(W_{old}) = \frac{1}{N} \Sigma (\hat{y} - y')$$

$$H(J(W_{old})) = \begin{bmatrix} \frac{\partial J^2}{\partial W_0^2} & \frac{\partial J^2}{\partial W_0 \partial W_1} & \frac{\partial J^2}{\partial W_0 \partial W_2} \\ \frac{\partial J^2}{\partial W_1 \partial W_0} & \frac{\partial J^2}{\partial W_1^2} & \frac{\partial J^2}{\partial W_1 \partial W_2} \\ \frac{\partial J^2}{\partial W_2 \partial W_0} & \frac{\partial J^2}{\partial W_2 \partial W_1} & \frac{\partial J^2}{\partial W_2^2} \end{bmatrix}$$

$$\frac{\partial J^2}{\partial W_0^2} = \frac{\partial \frac{1}{N} \Sigma (\hat{y} - y')}{\partial W_0} = \frac{1}{N} \Sigma \hat{y} (1 - \hat{y})$$

$$\frac{\partial J^2}{\partial W_0 \partial W_1} = \frac{1}{N} \Sigma \hat{y} (1 - \hat{y}) x$$

$$\frac{\partial J^2}{\partial W_0 \partial W_2} = \frac{1}{N} \Sigma \hat{y} (1 - \hat{y}) y$$

$$H(J(W)) = \frac{1}{N} \begin{bmatrix} \Sigma \hat{y}(1-\hat{y}) & \Sigma \hat{y}(1-\hat{y})x & \Sigma \hat{y}(1-\hat{y})y \\ \Sigma \hat{y}(1-\hat{y})x & \Sigma \hat{y}(1-\hat{y})x^2 & \Sigma \hat{y}(1-\hat{y})xy \\ \Sigma \hat{y}(1-\hat{y})y & \Sigma \hat{y}(1-\hat{y})yx & \Sigma \hat{y}(1-\hat{y})y^2 \end{bmatrix}$$

$$= \frac{1}{N} \Sigma \hat{y}_i (1 - \hat{y}_i) \begin{bmatrix} \Sigma 1 & \Sigma x & \Sigma y \\ \Sigma x & \Sigma x^2 & \Sigma xy \\ \Sigma y & \Sigma yx & \Sigma y^2 \end{bmatrix}$$

$$= \frac{1}{N} \sum \hat{y}_i (1-\hat{y}_i) \begin{bmatrix} 1 & 1 & \cdots & 1 \\ x_1 & \cdots & \cdots & x_N \\ y_1 & \cdots & \cdots & y_N \end{bmatrix} \begin{bmatrix} 1 & x_1 & y_1 \\ \vdots & \vdots & \vdots \\ 1 & x_N & y_N \end{bmatrix}$$

$$= \frac{1}{N} [\hat{y}_1 \cdots \hat{y}_N] \begin{bmatrix} 1-\hat{y}_1 \\ \vdots \\ 1-\hat{y}_N \end{bmatrix} \quad \cdots$$

。實際運算：

$$\Phi = \begin{bmatrix} 1 & \cdots & 1 \\ x_1 & \cdots & x_N \\ y_1 & \cdots & y_N \end{bmatrix}, \quad y' = \begin{bmatrix} y'_1 \\ \vdots \\ y'_N \end{bmatrix}, \quad w = \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix}$$

$$z = \Phi^T w, \quad \hat{y} = \sigma(z)$$

$$w' = \frac{1}{N} \Phi (\hat{y} - y')$$

$$H = \frac{1}{N} \hat{y}^T (1-\hat{y}) \Phi \Phi^T$$

$$w_{new} = w_{old} - \underbrace{H^{-1}}_{3\times 3} \cdot \underbrace{w'}_{3\times 1}$$

2. EM algorithm

$$X \quad \begin{bmatrix} 0 & 1 & \cdot & \cdot & \cdot \\ 0 & \cdot & & \cdot & \cdot \\ - & - & & & \cdot \\ \cdot & & & & 1 \\ 1 & & & & 1 \\ 1 & & & & \cdot \end{bmatrix} \quad \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \quad \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \quad \cdots$$

$$Z \qquad 1 \qquad\qquad 8 \qquad\qquad 9 \qquad \cdots$$

λ_k : chance of each images belong to C_k

$P[i][j][k]$: possibility that pixel $[i][j]$ of C_k appear 1

$w[a][k]$: responsibility that x_a belongs to C_k

$$w[a][k] = \frac{P(z_a = C_k, x_a \mid \theta)}{\sum_{i=1}^{9} P(z_a = C_i, x_a \mid \theta)}$$

$$P(z_a = C_k, x_a \mid \theta) = \lambda_k \prod_{i,j} P_{i,j,k}^{x_{a,i,j}} (1 - P_{i,j,k})^{(1 - x_{a,i,j})}$$

Likelihood:

$$L(\theta) = \prod_a \prod_k \left[\lambda_k \prod_{i,j} P_{i,j,k}^{x_{a,i,j}} (1 - P_{i,j,k})^{(1 - x_{a,i,j})} \right]^{w_{a,k}}$$

$$l(\theta) = \sum_a \sum_k w_{ak} \log \lambda_k \prod_{i,j} P_{ijk}^{x_{aij}} (1 - P_{ijk})^{(1 - x_{aij})}$$

$$= \sum_a \sum_k \left(w_{ak} \log \lambda_k + w_{ak} \sum_{i,j} \log \left(P_{ijk}^{x_{aij}} (1 - P_{ijk})^{(1 - x_{aij})} \right) \right)$$

$$= \sum_a \sum_k \left[w_{ak} \log \lambda_k + w_{ak} \sum_{i,j} x_{aij} \log P_{ijk} + w_{ak} \sum_{i,j} (1 - x_{aij}) \log (1 - P_{ijk}) \right]$$

$$\frac{\partial \ell}{\partial \lambda_k} = \sum_a \left(\frac{W_{ak}}{\lambda_k} - \frac{1-W_{ak}}{1-\lambda_k} \right) = 0$$

$$\Rightarrow \frac{\sum_a W_{ak}}{\lambda_k} = \frac{\sum_a 1-W_{ak}}{1-\lambda_k} \Rightarrow \sum_a W_{ak} - \lambda_k \sum_a W_{ak} = N\lambda_k - \lambda_k \sum_a W_{ak}$$

$$\Rightarrow \lambda_k = \frac{\sum_a W_{ak}}{N}$$

$$\frac{\partial \ell}{\partial P_{ijk}} = \sum_a \left(\frac{W_{ak} x_{aij}}{P_{ijk}} - \frac{W_{ak}(1-x_{aij})}{1-P_{ijk}} \right) = 0$$

$$\Rightarrow \frac{\sum_a W_{ak} x_{aij}}{P_{ijk}} = \frac{\sum_a W_{ak} - W_{ak} x_{aij}}{1-P_{ijk}} \Rightarrow \sum_a W_{ak} x_{aij} (1-P_{ijk}) = P_{ijk} \sum_a (W_{ak} - W_{ak} x_{aij})$$

$$\Rightarrow \sum_a W_{ak} x_{aij} = P_{ijk} \sum_a W_{ak}$$

$$\Rightarrow P_{ijk} = \frac{\sum W_{ak} x_{aij}}{\sum W_{ak}}$$

- 運算加速：向量化

P[i][j][k] ⇒ P[pixel][k] , 784 × 10

x[t][i][j] ⇒ x[t][pixel] , 60000 × 784

$$W_{t,k} = \frac{P(z_t = C_k, x_t \mid \theta)}{\sum_{k'=0}^{9} P(z_t = C_{k'}, x_t \mid \theta)} = \frac{\lambda_k \prod_i P_{ik}^{x_{ti}} (1-P_{ik})^{(1-x_{ti})}}{\sum_{k'=0}^{9} \lambda_{k'} \prod_i P_{ik'}^{x_{ti}} (1-P_{ik'})^{(1-x_{ti})}}$$

$$\log W_{tk} = \log \lambda_k + \sum_i \left[x_{ti} \log P_{ik} + (1-x_{ti}) \log (1-P_{ik}) \right]$$

$$- \sum_{k'} \left[\log \lambda_{k'} + \sum_i \left(x_{ti} \log P_{ik'} + (1-x_{ti}) \log (1-P_{ik}) \right) \right]$$

$$= \log \begin{bmatrix} \lambda_0 \\ \lambda_1 \\ \vdots \\ \lambda_9 \end{bmatrix} + X @ \log P_{ik} + (1-x_{ti}) @ \log (1-P_{ik}) - \sum$$

$$P_{ik} = \frac{\sum_t W_{tk} x_{ti}}{\sum_t W_{tk}} = X^T @ W \Big/ \sum_t W_{tk}$$

3. Mathematical Derivation

$$k^{(0)} = 0.5, \quad P_0^{(0)} = 0.6, \quad P_1^{(0)} = 0.1$$

$$X: \{ HHH, HHT, TTT \}$$

$$Z: \{ \quad \}$$

$W: \{ \quad \}$, W_i means chance X_i belg to C_1

$$P(Z_i = C_0, X_i \mid \theta) = k \binom{n}{X_i} P_0^{X_i} (1-P_0)^{n-X_i}$$

$$P(Z_i = C_1, X_i \mid \theta) = (1-k) \binom{n}{X_i} P_1^{X_i} (1-P_1)^{n-X_i}$$

$$L(\theta) = \prod \left[k \binom{n}{X_i} P_0^{X_i} (1-P_0)^{n-X_i} \right]^{(1-W_i)} \left[(1-k) \binom{n}{X_i} P_1^{X_i} (1-P_1)^{n-X_i} \right]^{W_i}$$

$$l(\theta) = \sum \left[(1-W_i) \log k \binom{n}{X_i} P_0^{X_i} (1-P_0)^{n-X_i} + W_i \log (1-k) \binom{n}{X_i} P_1^{X_i} (1-P_1)^{n-X_i} \right]$$

$$\frac{\partial l}{\partial k} = \sum \left(\frac{1-W_i}{k} - \frac{W_i}{1-k} \right) = 0$$

$$\Rightarrow \sum \frac{1-W_i}{k} = \sum \frac{W_i}{1-k} \quad \Rightarrow (1-k) \sum (1-W_i) = k \sum W_i$$

$$\Rightarrow n - nk - \sum W_i + k \sum W_i = k \sum W_i$$

$$\Rightarrow k = \frac{n - \sum W_i}{n} = 1 - \frac{\sum W_i}{n}$$

$$\frac{\partial l}{\partial P_0} = \frac{\partial}{\partial P_0} \sum \left[(1-W_i) \log P_0^{X_i} + (1-W_i) \log (1-P_0)^{n-X_i} \right]$$

$$= \sum \left(\frac{(1-W_i) X_i}{P_0} - \frac{(1-W_i)(n-X_i)}{(1-P_0)} \right] = 0$$

$$\Rightarrow \frac{\sum(1-W_i)x_i}{P_0} = \frac{\sum(1-W_i)(n-x_i)}{1-P_0}$$

$$\Rightarrow (1-P_0)\left(\sum x_i - \sum x_i W_i\right) = P_0\left(\sum n - \sum n W_i - \sum x_i + \sum x_i W_i\right)$$

$$\Rightarrow \sum x_i - \sum x_i W_i - P_0\sum x_i + P_0\sum x_i W_i = P_0\sum n - P_0\sum n W_i - P_0\sum x_i + P_0\sum x_i W_i$$

$$\Rightarrow \sum x_i - \sum x_i W_i = P_0\sum n - P_0\sum n W_i$$

$$\Rightarrow P_0 = \frac{\sum x_i - \sum x_i W_i}{\sum n - \sum n W_i} = \frac{\sum x_i(1-W_i)}{n\sum(1-W_i)}$$

$$\frac{\partial l}{\partial P_1} = \frac{\partial}{\partial P_1}\sum W_i \log P_1^{x_i}(1-P_1)^{n-x_i}$$

$$= \sum\left(\frac{W_i x_i}{P_1} - \frac{W_i(n-x_i)}{1-P_1}\right) = 0$$

$$\Rightarrow \frac{\sum W_i x_i}{P_1} = \frac{\sum W_i(n-x_i)}{1-P_1}$$

$$\Rightarrow (1-P_1)\sum W_i x_i = P_1\sum n W_i - P_1\sum W_i x_i$$

$$\Rightarrow \sum W_i x_i - P_1\sum W_i x_i = P_1\sum n W_i - P_1\sum W_i x_i$$

$$\Rightarrow P_1 = \frac{\sum W_i x_i}{n\sum W_i}$$

- E step: $\qquad X: \{HHH, HHT, TTT\}$

$k^{(0)} = 0.5, \; P_0^{(0)} = 0.6, \; P_1^{(0)} = 0.1$

$$W_i = \frac{P(Z_i = C_1, X_i \mid \theta)}{P(Z_i = C_0, X_i \mid \theta) + P(Z_i = C_1, X_i \mid \theta)}$$

X_0 : HHH

$$W_0 = \frac{k\binom{3}{3}P_0^3}{k\binom{3}{3}P_0^3 + (1-k)\binom{3}{3}P_1^3} = \frac{0.5\times0.6^3}{0.5\times0.6^3 + 0.5\times0.1^3}$$

$$= \frac{1.08}{1.08 + 0.0005} \doteqdot 0.9995$$

X_1 : HHT

$$W_1 = \frac{k\binom{3}{2}P_0^2(1-P_0)^1}{k\binom{3}{2}P_0^2(1-P_0)^1 + (1-k)\binom{3}{2}P_1^2(1-P_1)^1}$$

$$= \frac{0.5\times3\times0.6^2\times0.4}{0.5\times3\times0.6^2\times0.4 + 0.5\times3\times0.1^2\times0.9} = \frac{0.216}{0.216+0.0135} \doteqdot 0.9412$$

X_2 : TTT

$$W_2 = \frac{k\binom{3}{0}(1-P_0)^3}{k\binom{3}{0}(1-P_0)^3 + (1-k)\binom{3}{0}(1-P_0)^3}$$

$$= \frac{0.5\times0.4^3}{0.5\times0.4^3 + 0.5\times0.9^3} = \frac{0.032}{0.032+0.3645} \doteqdot 0.0807$$

- M step:

$$k^{(1)} = 1 - \frac{\sum W_i}{n} = 1 - \frac{0.9995 + 0.9412 + 0.0807}{3} = 0.3262$$

$$P_0^{(1)} = \frac{\sum x_i(1-W_i)}{n\sum(1-W_i)} = \frac{3\times 0.0005 + 2\times 0.0588 + 0\times 0.9193}{3\times \quad 0.3262\times 3} \doteq 0.0406$$

$$P_1^{(1)} = \frac{\sum W_i x_i}{n\sum W_i} = \frac{3\times 0.9995 + 2\times 0.9412 + 0\times 0.0807}{3\times(0.9995 + 0.9412 + 0.0807)} \doteq 0.8049$$